

KAMARAJ PORTAL – WEB-BASED CODING ASSESSMENT PLATFORM

Introduction

Kamaraj Portal is a web-based coding assessment and learning platform developed to provide students with a centralized environment for practicing programming problems, submitting solutions, and monitoring their performance. The platform is designed specifically for academic use, allowing faculty to conduct programming activities and students to participate in coding exercises through a simple and organized web interface. It complements the college's existing digital ecosystem, which provides various academic and student-related resources.

Objectives

The main objective of Kamaraj Portal is to simplify the process of conducting programming exercises and evaluating student performance. The system reduces the need for manual correction by automatically executing submitted programs and checking the output against predefined test cases. It also encourages students to improve their problem-solving and programming skills through regular practice and competitive scoring.

Major Features

The portal provides secure student login and logout, allowing each student to access their individual account. The Dashboard displays the current coding question, previous questions, submission information, and other relevant activities. Students can write their solutions and submit programs directly through the Submit page.

A major feature of the platform is the automatic code evaluation system. Submitted programs are executed in an isolated environment and their outputs are compared with the expected results. The system supports multiple programming languages such as Python, Java, C++, and JavaScript. A points and penalty system is also implemented to encourage students to solve problems efficiently. Students receive a base score for successful submissions, while additional penalties can be applied based on the submission time.

The portal also includes a Leaderboard, which displays student rankings based on their scores. This creates a competitive learning environment and motivates students to participate actively in coding activities.

Technology Used

The initial prototype of the system was developed using Python Flask for the backend, PostgreSQL for persistent data storage. NsJail based isolation can be used to execute student programs safely and prevent submitted programs from affecting the main server.

Benefits

Kamaraj Portal provides a centralized and automated solution for programming assessments. It reduces faculty workload, provides faster evaluation, maintains student performance records, and allows students to receive immediate feedback on their submissions. The leaderboard and scoring mechanism further improve student engagement and encourage healthy competition.

Conclusion

Kamaraj Portal is an academic coding platform designed to improve programming practice and assessment within the college environment. By combining web technologies, database management, automated code execution, and performance tracking, the system provides a scalable foundation for conducting coding exercises and online programming assessments. The platform can be further enhanced with features such as additional programming languages, advanced analytics, automated test-case generation, contests, notifications, and stronger sandbox isolation.